

Listening to Silence: Attention Sinks, Registers, and the Hidden Work of Padding in Speech Transformers

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Whisper, the most widely deployed speech recognizer and the audio encoder inside many audio language models, always pads its input to 30s, and it fails when the padding is removed: in our pilot, word error rate (WER) on short LibriSpeech utterances rises from 5.3% to 373% for Whisper-base and from 2.0% to 428% for Whisper-large-v3-turbo, mostly through runaway repetition. Existing fixes retrain the model or a learned input segment, and the role of the silent frames is not understood. We study what speech transformers do with silence through the lens of attention sinks, massive activations and register tokens—phenomena with competing 2026 theories that have been examined in language and vision models but not in speech encoders. In three Whisper sizes the encoder forms textbook high-norm, massive-activation tokens on the padding, yet these tokens are dispensable for recognition: masking or removing them leaves WER unchanged. What Whisper needs is silence used in two ways: decoder cross-attention to silent encoder states as end-of-speech evidence, and encoder self-attention to a few position-anchored sink frames at the edges of the 30s window. In large-v3-turbo, 16 cached high-attention silence frames give 2.6% WER versus 17.5% for 16 random ones. Building on this, SILENCECACHE precomputes silence keys, values and outputs once and encodes only the speech; it matches or improves full-padding WER (5.1/2.6/1.7% vs. 5.3/2.6/2.0%) at 0.16× the encoder FLOPs, without any training. These numbers come from a 100-utterance pilot; the full evaluation is specified in Sec. V.

I. INTRODUCTION

Whisper [1] processes every input as a 30s window. A two-second voice command is zero-padded with 28s of digital silence before the encoder runs, so roughly 90% of the encoder computation is spent on frames that carry no signal. Removing the padding is the obvious fix, and it fails badly. On 100 short LibriSpeech [2] utterances (2–8s), feeding only the speech frames raises greedy-decoding WER from 5.3% to 373% for Whisper-base, from 2.6% to 214% for Whisper-small and from 2.0% to 428% for Whisper-large-v3-turbo; most outputs loop on the last phrase (Table II). Because Whisper’s encoder is also the audio front-end of many audio language models, the cost of the padding, and of not understanding it, propagates widely.

Current remedies treat the padding as an engineering obstacle. WhisperFlow trains a 0.5s “hush word” appended to the audio [3]; WhisperKit self-distills the encoder with block-diagonal masks so that a silent block can be cached [4]; other systems detect and truncate hallucinated continuations online [5] or drop encoder tokens [6]. Each of these either retrains something or works around the failure without explaining it. The basic question remains open: *what does a speech transformer compute with 28s of silence?*

A natural hypothesis comes from the literature on attention sinks, massive activations and register tokens. Vision transformers recruit uninformative background patches as high-norm “registers” that store global information [7, 8]; language models dump attention on

the first token, whose massive activations act as implicit biases [9, 10]. These phenomena are among the most actively studied transformer mechanisms in 2026, with competing accounts of why they arise and whether they matter: pre-normalisation makes sinks and massive activations co-occur but each can exist without the other [11]; sinks stem from the unaggregated first token under causal masking [12]; softmax attention needs a sink to implement a default “no-op” state [13]; spikes land on the least-shared token [14]; or high-norm artifacts are a symptom of lazy aggregation rather than the cause [15]. None of this work examines speech encoders, where “nothing” is not a rare background patch but a large, exactly controllable part of every input.

We examine Whisper through this lens and find a sharper picture than the register hypothesis predicts. Whisper’s encoder does form textbook registers on the padding: a few frames reach 29–59× the median token norm, and in Whisper-base and -small they carry massive activations above Sun et al.’s threshold [9] (Whisper-base: a single activation of 501, 1,122× the median magnitude). But they are *dispensable*: masking encoder attention to them, or removing them from the decoder’s view, leaves WER unchanged in all three models. What Whisper actually needs is *silence*, used in two ways. First, the decoder needs silent encoder states as end-of-speech evidence; without them it loops. Second, the encoder’s speech frames attend to a handful of *attention-sink* frames that sit at fixed positions near the start and end of the 30s window. These two roles can be supplied without encoding any padding: a content-free silence input is encoded once, its keys, values and outputs are cached, and each utterance is encoded alone while attending to the cache. We call this SILENCECACHE. In the pilot it matches or improves full-padding WER at 0.16× the encoder FLOPs,

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and its reliance on the sink frames grows with model size: for large-v3-turbo, caching the 16 most-attended silence frames recovers 2.6% WER, while 16 random silence frames give 17.5% (Fig. 3).

a. Contributions. (i) A protocol and census of registers, massive activations and attention sinks in speech and audio encoders, with natural experiments over normalisation type, special tokens and padding regime (visible in Whisper, masked in Whisper-derived audio-LLM encoders, absent in windowed encoders). (ii) Controlled acoustic interventions that pit the 2026 accounts of sink formation against each other by varying the energy, redundancy, position and duration of non-speech input. (iii) Causal evidence, in production speech models, that functional attention sinks and high-norm registers dissociate, together with an anatomy of how the decoder uses silence to stop. (iv) SILENCECACHE, a training-free method for padding-free Whisper that follows from this mechanism, benchmarked against trained alternatives. Items (i)–(iii) are supported by pilots reported here; the full-scale evaluation is specified in Sec. V, and the evidence log behind every claim is maintained in the project repository.

II. BACKGROUND

A. Problem Setting

a. Whisper’s input window. An utterance x of duration $\tau \leq 30$ s is converted to an 80- or 128-channel log-mel spectrogram with a 10 ms hop and zero-padded to 3000 frames; a two-layer convolutional stem (second layer stride 2) yields $N = 1500$ encoder positions, to which fixed sinusoidal position embeddings $\mathbf{p}_1, \dots, \mathbf{p}_N$ are added [1]. The encoder is a stack of L pre-normalisation (pre-LN) transformer blocks followed by a final LayerNorm; every Whisper size uses 64-dimensional attention heads (tiny: 4 layers, width 384; large: 32 layers, width 1280). Let $T = \lceil 50\tau \rceil$ be the number of positions covered by speech. With padding, positions $T+1, \dots, N$ hold the encoding of digital silence; the decoder cross-attends to all N encoder outputs $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_N]$.

b. Goal. We ask which computations the padded positions perform, and whether those computations can be supplied when the encoder processes only the T speech positions. The task performance measure is the word error rate of the decoded transcript,

$$\text{WER} = \frac{S + D + I}{N_{\text{ref}}}, \quad (1)$$

with substitutions S , deletions D and insertions I against N_{ref} reference words after Whisper’s text normaliser. Because insertions are unbounded, WER exceeds 100% when the decoder loops; we therefore also report a hallucination rate (Sec. VB). All interventions are applied at inference time to frozen public checkpoints; no parameter is trained.

B. Notation and Preliminaries

For a layer ℓ with hidden states $\mathbf{X}^{(\ell)} \in \mathbb{R}^{N \times d}$, a head with queries $\mathbf{q}_i$, keys $\mathbf{k}_j$ and values $\mathbf{v}_j$ computes

$$\mathbf{o}_i = \sum_j a_{ij} \mathbf{v}_j, \quad a_{ij} = \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j)}{\sum_m \exp(\mathbf{q}_i^\top \mathbf{k}_m)}. \quad (2)$$

We use three observables that recent work shows can dissociate [11, 14].

Definition 1 (Register frame, massive activation, attention)

A register (high-norm) frame at layer ℓ is a position whose hidden-state norm exceeds ten times the median token norm, $\|\mathbf{x}_j^{(\ell)}\|_2 > 10 \text{ med}_m \|\mathbf{x}_m^{(\ell)}\|_2$. A massive activation is a single entry with $|x_{jc}^{(\ell)}| > 100$ that is at least about $1000\times$ the median entry magnitude of the layer [9]; we report the ratio $\Psi^{(\ell)} = \max_{j,c} |x_{jc}| / \text{med}_{j,c} |x_{jc}|$. An attention-sink frame is a position whose received attention, averaged over heads and queries, $\bar{a}_j = \frac{1}{HN} \sum_{h,i} a_{ij}^h$, is among the largest in the layer; following Gu et al. [16], a sink holds at least 30% of the attention mass in some layer.

The following elementary fact explains why a sink can implement a “no-op” and why sink keys and values can be cached.

Proposition 1 Fix a head and a query $\mathbf{q}_i$. Suppose one key $\mathbf{k}_s$ satisfies $\mathbf{q}_i^\top \mathbf{k}_s - \max_{j \neq s} \mathbf{q}_i^\top \mathbf{k}_j \geq \Delta$ and all values are bounded, $\|\mathbf{v}_j\|_2 \leq V$. Then

$$\|\mathbf{o}_i - \mathbf{v}_s\|_2 \leq 2V(n-1)e^{-\Delta}, \quad (3)$$

where n is the number of keys. Hence, as the logit gap grows, the head output tends to the constant $\mathbf{v}_s$, independent of the content of the other tokens.

Proof. $1 - a_{is} = \sum_{j \neq s} a_{ij} \leq (n-1)e^{-\Delta}$ and $\mathbf{o}_i - \mathbf{v}_s = \sum_{j \neq s} a_{ij}(\mathbf{v}_j - \mathbf{v}_s)$, whose norm is at most $2V(1 - a_{is})$. $\square$

Proposition 1 is the reason a sink acts as an attention bias [8, 9]: what matters for the other tokens is the sink’s key and value, not where they were computed. If the keys and values of the silent positions do not depend on the utterance, they can be computed once and reused. Sec. IV tests how far this holds for Whisper.

III. RELATED WORK

A. Closest Methodological Line

a. Registers, sinks and massive activations. High-norm artifact tokens in vision transformers were linked to redundant background patches and global information and removed by training with extra register tokens [7]. Test-time registers move them into an added token by editing a sparse set of “register neurons” without training [8],

and RegCache prefixes cached register keys and values to suppress activation outliers for quantization [17]. In language models, attention sinks on the first token enable streaming [10], and massive activations act as fixed biases [9]. The 2026 literature disputes the mechanism: pre-normalisation couples sinks and massive activations but either can be removed without loss [11]; sinks arise from the first token’s unaggregated variance under causal masking [12]; massive activations emerge in a single layer [18] and cause compression valleys [19]; sinks are provably needed for default states [13]; outlier-driven rescaling is essential for training [20]; and gating removes sinks [21]. In vision, artifacts are attributed to lazy aggregation [15], and high-norm tokens were not found in bidirectional text encoders [22]. All of this work concerns text, images or diffusion models; none studies speech encoders, silence or fixed-length padding.

b. Sinks at the encoder-LLM interface. High-norm ViT tokens propagate into vision-language models and carry global semantics [23]; visual massive activations in LVLMs are brittle and follow a least-shared location rule [14]. For audio, sinks have been characterised only inside the language model of audio-visual LLMs [24–26], and audio-token attention has been predicted from encoder outputs for pruning [27]. The audio encoder itself, and in particular its silent frames, is not analysed in any of these works.

B. Complementary Research Direction

a. Efficient and padding-free Whisper. Short inputs make Whisper hallucinate, so streaming systems pad and overlap buffers. WhisperFlow appends a trained 0.5 s audio “hush word” [3]; WhisperKit self-distills block-diagonal attention so that a silent block’s encoder output can be cached [4]; NPUSper stops decoding when cross-attention moves backwards [5]; stride- k subsampling drops encoder tokens without training [6]; Moonshine removes padding by retraining with rotary positions [28]. Separately, a line of work mitigates Whisper’s hallucinations on non-speech audio by masking heads [29] or steering encoder representations [30]. These methods differ from ours in information access and cost: they require training data and optimisation, or address a different failure (non-speech input, decoding control), and none offers a mechanistic account of why padding is needed.

C. Positioning of This Work

The closest methods are test-time registers [8] and RegCache [17], which add or cache register tokens in vision encoders, and WhisperKit’s silence caching [4], which reuses a silent block after retraining with block masks. We differ in three ways. First, we study speech encoders and their decoders, where the “uninformative” input is silence that can be manipulated exactly. Second,

we separate functional attention sinks from high-norm registers in production models and find that only the former matter for recognition. Third, SILENCECACHE caches silence without retraining and without block masks: speech queries attend to cached silence keys and values, and the decoder sees a cached silence tail.

IV. METHOD

A. Overview

Our study has two parts. The *measurement and intervention protocol* records, for any frozen speech encoder, per-layer norms, activation magnitudes and received attention, labels each position as speech, in-utterance silence, padding or a fixed absolute position, and applies causal edits: masking attention to selected positions, removing positions from the decoder’s view, and replacing computed keys, values and outputs with cached ones. The *method*, SILENCECACHE, uses one of these edits as an inference procedure: it encodes a 30 s silence input once, caches its per-layer keys and values and its final outputs, and then encodes each utterance without padding while letting speech queries attend to the cached silence and letting the decoder read a cached silence tail. The protocol is new in that it targets silence; SILENCECACHE is new in that it removes padding without training. Attention, normalisation and decoding are inherited unchanged from Whisper [1].

a. Cached keys and values. Let $\mathbf{K}_{\emptyset}^{(\ell)}, \mathbf{V}_{\emptyset}^{(\ell)} \in \mathbb{R}^{N \times d}$ be the keys and values of layer ℓ and $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_N]$ the encoder outputs when the input is pure digital silence. For an utterance covering T positions and a set $\mathcal{C} \subseteq \{T+1, \dots, N\}$ of cached positions, each speech query $\mathbf{q}_i$ ($i \leq T$) attends over

$$\mathbf{K}_i^{(\ell)} = [\mathbf{K}_{1:T}^{(\ell)}; \mathbf{K}_{\emptyset, \mathcal{C}}^{(\ell)}], \quad (4)$$

$$\mathbf{V}_i^{(\ell)} = [\mathbf{V}_{1:T}^{(\ell)}; \mathbf{V}_{\emptyset, \mathcal{C}}^{(\ell)}], \quad (5)$$

and the decoder cross-attends to $[\mathbf{H}_{1:T}; \mathbf{S}_{\mathcal{D}}]$ for a set $\mathcal{D}$ of tail positions. With $\mathcal{C} = \mathcal{D} = \{T+1, \dots, N\}$ the only approximation relative to padded encoding is that the cached silence states were computed without the utterance in context (padding positions do not attend to speech). Restricting $\mathcal{C}$ to the k silence positions with the largest received attention $\bar{a}_j$ in the silence-only run gives SILENCECACHE-top- k ; drawing $\mathcal{C}$ uniformly at random gives the matched control.

B. Framework Overview

Figure 1 shows the inference path. The diagnostic variants used in Sec. V switch individual arrows off: *tail only* ($\mathcal{C} = \emptyset$), *KV only* ($\mathcal{D} = \emptyset$), top- k versus random- k caches, and a decoder that sees only the top- k tail

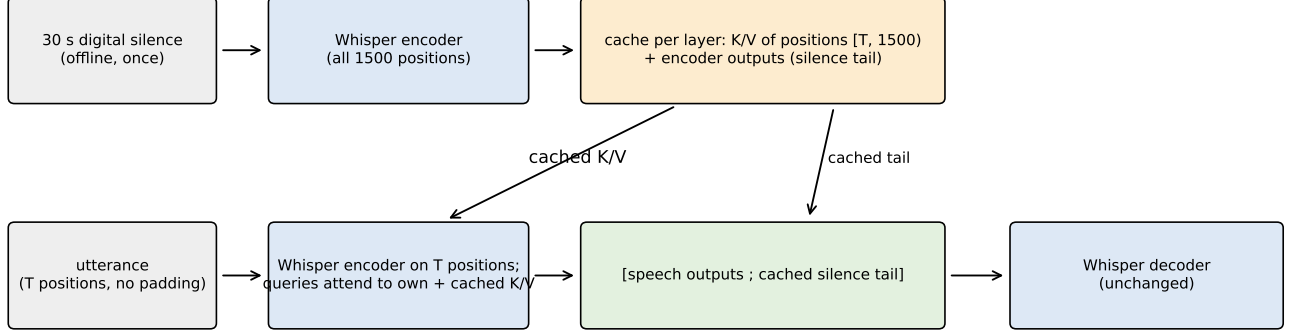


FIG. 1. SILENCECACHE removes Whisper’s padding without training. Offline (top), a 30s silence input is encoded once and its per-layer keys/values for positions $[T, 1500]$ and its encoder outputs (the “tail”) are cached. Online (bottom), only the T speech positions are encoded; speech queries attend to their own keys plus the cached silence keys (all, or the top- k attention sinks), and the unchanged decoder cross-attends to the speech states followed by the cached tail. Nothing is trained; the cost scales with T rather than 1500 positions.

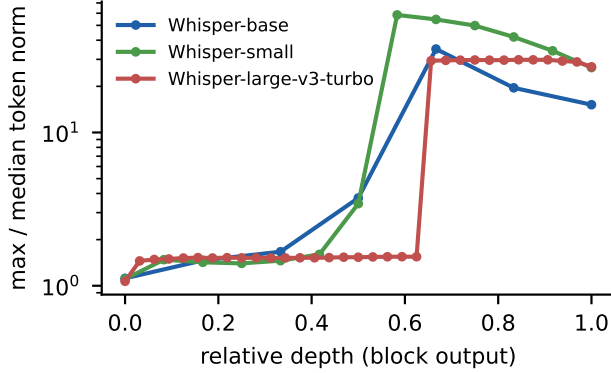


FIG. 2. Whisper’s encoder forms high-norm register frames on silence. Max/median token norm per block output for a 30s silence input. The ratio jumps abruptly at one block (base: block 4 of 6; small: 7 of 12; large-v3-turbo: 20 of 32) and stays high, the single-layer emergence pattern reported for language models [18].

positions. Together they separate the encoder-side and decoder-side roles of silence.

C. Core Components

a. Faithful forward pass. We re-implement the Whisper encoder forward pass explicitly (convolutional stem, sinusoidal positions sliced to the input length, pre-LN blocks, final LayerNorm), so that inputs of any length and extra cached keys/values can be used. On a padded LibriSpeech utterance its output differs from the Hugging Face implementation by at most 6.9×10^{-4} (output scale 23.6), and greedy decoding produces identical transcripts. Hidden states are recorded *before* the final LayerNorm,

because library conventions for returned hidden states differ across models and versions.

b. Observables. Per layer we record the norm ratio, $\Psi^{(\ell)}$ and the absolute maximum activation, received attention per position, and, in the planned census, matrix entropy and anisotropy of $\mathbf{X}^{(\ell)}$ [19]. Positions are labelled with an energy voice-activity threshold (-40 dBFS), with the padding boundary, and with absolute index.

c. Interventions. (a) *Encoder masking*: set the logits of selected key positions to $-\infty$ from a chosen block onwards. (b) *Decoder removal*: delete selected positions from the encoder output before cross-attention. (c) *Caching*: replace computed silence with cached keys, values and outputs as in Eq. (5). Selections are either by norm (top- k register frames at the emergence block) or by attention (top- k sink frames in the silence-only run), with random- k controls.

D. Algorithm

Algorithm 1: SILENCECACHE inference

Require: Whisper encoder/decoder; utterance x ; cache size k (or “all”)

- 1: **Offline, once:** encode 30 s of silence; store $\mathbf{K}_{\emptyset}^{(\ell)}, \mathbf{V}_{\emptyset}^{(\ell)}$ for all layers, outputs $\mathbf{S}$, and received attention $\bar{a}_j$
- 2: Compute log-mel of x without padding; $T \leftarrow$ number of encoder positions
- 3: $\mathcal{C} \leftarrow$ top- k of $\{\bar{a}_j\}_{j>T}$ (or all $j > T$)
- 4: **for** $\ell = 1, \dots, L$ **do**
- 5: compute $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ for the T speech positions
- 6: attend with keys/values $[\mathbf{K}; \mathbf{K}_{\emptyset, \mathcal{C}}^{(\ell)}], [\mathbf{V}; \mathbf{V}_{\emptyset, \mathcal{C}}^{(\ell)}]$; MLP on T positions
- 7: **end for**
- 8: $\mathbf{H} \leftarrow$ final LayerNorm output for the T positions
- 9: **return** decode($[\mathbf{H}; \mathbf{S}_{T+1:N}]$)

E. Complexity or Theoretical Analysis

Counting multiply-accumulates as two FLOPs, one encoder layer with width d , MLP width $f = 4d$, T_q query positions and T_k key positions costs

$$C(T_q, T_k) = 2T_q(4d^2 + 2df) + 4T_qT_kd. \quad (6)$$

Padded inference costs $C(N, N)$ per layer. SILENCECACHE with a full cache costs $C(T, N)$ and with a top- k cache $C(T, T+k)$. For Whisper-base ($d = 512$) and a 4.7 s utterance ($T \approx 234$), $C(T, N)/C(N, N) \approx 0.16$ and $C(T, T+16)/C(N, N) \approx 0.11$, essentially the cost of no padding. Memory for the cache is $2Lkd$ values ($k = 16$, large-v3: 1.3 M values). Wall-clock speed-ups depend on kernels and batch size and are measured separately (Sec. V A); the derivation of Eq. (6) is in §B.

V. EVALUATION

We organise the evaluation around four research questions: **RQ1** where registers, massive activations and sinks form in speech encoders; **RQ2** which account of sink formation predicts their location; **RQ3** which of these structures Whisper needs, in the encoder and in the decoder; **RQ4** whether SILENCECACHE removes padding without loss. This draft reports the *pilot* that motivated the study (three Whisper sizes, 100 utterances each) and specifies the full evaluation. Pilot numbers are marked as such and should be read as indicative.

A. Experimental Setup

The pilot uses the 100 LibriSpeech test-clean utterances with duration 2–8 s selected by a fixed random

seed (mean 4.7 s), greedy decoding with the prompt `<|startoftranscript|><|en|><|transcribe|><|notimestamps|>` a 160-token limit, and frozen public checkpoints. The full evaluation (planned) extends this to six Whisper sizes plus distil-large-v3, LibriSpeech test-clean/other in full, TED-LIUM 3, FLEURS in seven languages including Bengali, a short-command set, and noisy and long-form conditions; it adds wall-clock timing on T4 and RTX-3050 GPUs at batch sizes 1 and 16. The census for RQ1 covers about 18 checkpoints: Whisper, Moonshine (trained without padding [28]), post-LN and pre-LN self-supervised encoders, AST with its CLS and distillation tokens [31], and the audio towers of audio language models that mask padding or use windows. Implementation details are listed in §C.

TABLE I. Pilot configuration. Every value is fixed a priori; nothing is tuned on the test utterances.

Parameter	Setting
Dataset	LibriSpeech test-clean, 100 utts, 2–8 s
Models	Whisper base, small, large-v3-turbo
Training	none (frozen checkpoints)
Decoding	greedy, English, no timestamps
Cache source	one 30 s digital-silence input
Random seeds	0 (utterances, random- k)
Hardware	RTX 3050 Laptop 4 GB (fp32; turbo fp16)

B. Evaluation Metrics

WER follows Eq. (1) after Whisper’s English normaliser. Because looping transcripts inflate WER without bound, we also report the hallucination rate

$$\text{Hal} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[|\hat{y}_i| > |y_i| + 3 \vee \text{rep}(\hat{y}_i)], \quad (7)$$

where $|\cdot|$ counts words and rep detects a three-word phrase repeated at least twice. Cost is reported as encoder FLOPs relative to padded inference (Eq. (6)); the full evaluation adds measured latency, real-time factor and peak memory, and 95% bootstrap confidence intervals with paired tests against padded inference.

C. Main Results

Table II compares padded inference, removal of padding, real partial padding and the SILENCECACHE variants on the same utterances. Three results stand out. First, removing padding is catastrophic for all sizes (WER 214–428%), and real 5 s padding is not a safe substitute: it is adequate for base and small but gives 35.1% for large-v3-turbo. Second, the two roles of silence are separable and complementary. The cached tail alone fixes most loops for base and small but not for turbo (22.7%), while

cached keys and values alone do not prevent loops; together they match or improve padded WER in every model (5.12/2.64/1.71% vs. 5.28/2.56/2.02%) at $0.157\times$ the encoder FLOPs. Third, removing high-norm register frames from the decoder’s view, or masking encoder attention to them from the emergence block onwards, leaves WER essentially unchanged.

The top- k results locate the encoder-side effect (Fig. 3). For base and small, the choice of cached positions barely matters, consistent with their weak dependence on encoder-side silence. For large-v3-turbo the gap is large: 16 high-attention positions give 2.64% WER versus 17.53% for 16 random ones, and the curves meet only at $k = 256$. The highest-attention silence positions are at fixed absolute indices: for large-v3-turbo they are positions 0, 1496, 1499, 1487, 1455, 1495, 1456 and 1494. This suggests why 5s of real padding fails for turbo: it never fills the end-of-window positions. We treat this as a hypothesis for the full evaluation (H3d in the project plan).

D. Ablation Study

Two ablations isolate what SILENCECACHE relies on. *Registers versus sinks.* Masking or removing the highest-norm frames, the registers in the sense of [7], does not change WER (Table II, last rows), whereas restricting the cache to random rather than high-attention positions does. Functional reliance therefore follows received attention, not norm. This agrees with evidence that sinks can exist without massive activations [11]. One caveat: in base, the high-attention silence positions include the top-norm frames (positions 1445, 1446, 1453 and 1454), so the overlap between the two sets must be measured per layer before the dissociation is claimed in general (planned experiment E4). *Decoder tail size.* With the full key/value cache, base and small still need many tail positions (14.27% and 12.34% WER with only the top-16), while turbo needs very few (1.94%). How much silence the decoder must see, and whether nearest-first or highest-attention tail positions matter, is the subject of a planned tail-length sweep.

E. Additional Analysis

a. Partial padding helps gradually, but only in small models. With real zero padding on Whisper-base, WER falls from 31.2% with 1s to 16.3% with 2s and 5.9% with 5s or 10s (§D), whereas large-v3-turbo still has 35.1% with 5s (Table II). Transplanting the keys/values of four register frames from a padded run into an unpadded one does not help (315% WER), while appending a padding tail from a *speech* reference gives 24.5%, far worse than the content-free silence tail (5.66%). The tail must therefore be silence-conditioned, not merely “padding-shaped”.

b. Non-speech audio. On 200 ESC-50 clips [32] the silence tail shortens runaway outputs but does not prevent short hallucinations. For large-v3-turbo the share of clips with any output is 78.5% with padding, 98.0% without, and 100% with the tail; mean output length is 1.10, 16.27 and 1.32 words respectively. Non-speech hallucination is therefore treated as exploratory (§D).

c. Planned analyses. (a) The RQ1 census with entropy and anisotropy per layer. (b) A factorial over the energy, redundancy, position and duration of inserted non-speech blocks, which pits the redundancy [7], least-shared [14] and self-sinking [11, 12] accounts against each other. (c) Mask-to-self interventions [12]. (d) Decoder cross-attention anatomy at end-of-transcript steps. (e) Audio-LLM encoders whose padding is masked or absent. (f) A benchmark against a re-trained hush word [3], fine-tuned short-context checkpoints, stride- k [6] and combinations.

VI. DISCUSSION

The pilot changes the question from “do speech transformers need registers?” to “what does silence do for them?”. Its answer is that Whisper uses silence as a source of attention sinks and end-of-speech evidence, and that the high-norm register tokens, though present, are not what it relies on. The causal language here is limited to interventions on three models and 100 utterances; the full evaluation is designed to test each statement at scale.

A. Interpretation

Two mechanisms are consistent with the results. On the decoder side, silent encoder states give cross-attention somewhere to go once the transcript is complete. Without them the decoder re-reads the speech and loops, which matches the observation that unpadded failures are dominated by continuations beyond the reference [5], and the view that softmax attention needs a sink to realise a default state [13]. On the encoder side, speech queries of larger models attend to a few silence positions at fixed indices near the window edges. Proposition 1 explains why such sinks can be cached: what matters to other tokens is the sink’s key and value. Their anchoring to absolute positions resembles the first-token sink of causal language models [11, 12], but here it appears at both ends of a bidirectional window. Why it appears there, and why it strengthens with scale, is open. The planned factorial and mask-to-self experiments address it. We label as speculative the idea that end-of-window sinks arise because padding always occupies those positions during training.

TABLE II. **Pilot: what Whisper needs from its padding** (LibriSpeech test-clean, 100 utterances of 2–8 s, greedy decoding). WER and hallucination rate (Hal, Eq. 7) in %, encoder FLOPs relative to 30 s padding. “tail”: the decoder also sees cached silence outputs; “KV”: speech queries attend to cached silence keys/values; top-/random-16: only 16 cached positions (largest received attention in the silence run vs. random). Shaded rows are SILENCECACHE. Register frames are the highest-norm positions at the emergence block. All numbers are pilot results; confidence intervals are part of the planned full evaluation.

Condition	Whisper-base			Whisper-small			Whisper-large-v3-turbo		
	WER	Hal	FLOPs	WER	Hal	FLOPs	WER	Hal	FLOPs
30 s padding (standard)	5.28	1.0	1.000	2.56	1.0	1.000	2.02	1.0	1.000
no padding	372.77	43.0	0.115	214.35	27.0	0.125	427.77	60.0	0.136
5 s real padding	5.82	1.0	0.253	2.79	1.0	0.271	35.07	13.0	0.288
cached tail only	5.66	1.0	0.115	3.96	1.0	0.125	22.73	0.0	0.136
cached KV only	61.60	10.0	0.157	426.84	48.0	0.157	316.52	72.0	0.157
SILENCECACHE (KV + tail)	5.12	1.0	0.157	2.64	1.0	0.157	1.71	1.0	0.157
SILENCECACHE-top-16 (KV + tail)	5.43	1.0	0.115	3.80	1.0	0.126	2.64	1.0	0.136
random-16 KV + tail (control)	5.74	1.0	0.115	3.96	1.0	0.126	17.53	0.0	0.136
KV + top-16 tail positions only	14.27	2.0	0.157	12.34	2.0	0.157	1.94	1.0	0.157
padded; 16 register frames removed from decoder	5.51	1.0	1.000	2.56	1.0	1.000	2.02	1.0	1.000
padded; encoder attention to 16 register frames masked	—	—	—	2.40	1.0	1.000	1.86	1.0	1.000

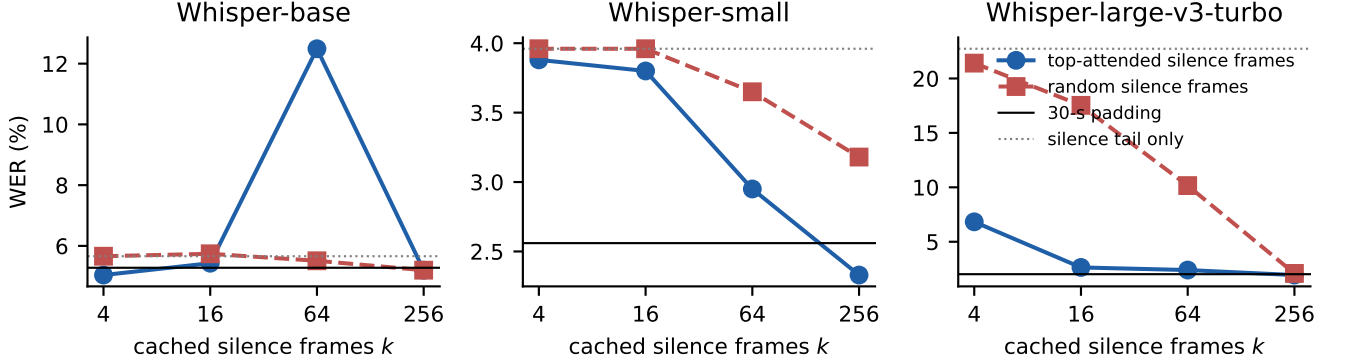


FIG. 3. The encoder’s use of silence is concentrated in a few attention-sink positions, and increasingly so with model size. WER when speech queries attend to k cached silence positions chosen by received attention in the silence-only run (blue) or at random (red); the decoder sees the full cached tail in both cases. For large-v3-turbo, 4–16 sink positions recover most of the gap to padded inference, while random positions do not. Pilot, 100 utterances.

B. Limitations

The pilot uses read English speech, short utterances, greedy decoding and three checkpoints. Long-form decoding with timestamps, beam search, noisy and multilingual speech are untested, and the large-v3 encoder was not yet run. FLOPs are analytic; attention to cached keys may not yield proportional wall-clock gains with all kernels. The cache is computed from digital silence; recordings whose “silence” is noise may need a cache matched to the noise floor. The dissociation between sinks and registers depends on how the two sets are selected; this is unresolved where they overlap (e.g. base). Finally, SILENCECACHE assumes access to model internals, unlike API-only deployments.

C. Broader Implications and Future Work

If the account holds, three things follow. Any transformer trained with fixed-length padding may learn to depend on it, so removing padding should preserve its

sinks rather than simply truncate. Analyses of speech representations should report silence handling, because massive activations and sinks on silent frames dominate norms and attention statistics. And audio language models built on Whisper encoders inherit, mask or lose these sinks depending on their padding regime, which may matter for audio-token pruning and hallucination. Future work includes the audio-LLM natural experiment, streaming with cached sinks, and training-time interventions (explicit silence registers or gated attention [21]) that would make speech models robust to variable-length input by design.

VII. CONCLUSION

Whisper fails without its 30 s padding, and we asked what the silence is used for. Pilot interventions on three Whisper sizes show that the high-norm register tokens the encoder forms on padding are dispensable, while silence serves two roles that are not: end-of-speech evidence for the decoder and a few position-anchored attention sinks for the encoder, whose importance grows with model size.

Caching these silence states once gives SILENCECACHE, a training-free way to encode only the speech, which matched or improved padded WER at about one sixth of the encoder FLOPs in the pilot. These conclusions rest on 100 utterances per model; the specified full-scale evaluation will determine how far they generalise across

sizes, languages, conditions and audio language models.

ACKNOWLEDGMENTS

To be completed. Pilot experiments ran on a single NVIDIA RTX 3050 Laptop GPU (4 GB); full-scale experiments are planned on Kaggle T4 GPUs.

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Appendix A: Notation and Symbols

This section collects the notation used in the paper. The WER is defined in Eq. (1), attention in Eq. (2), the cached keys/values of SILENCECACHE in Eq. (5), the hallucination rate in Eq. (7) and the per-layer cost in Eq. (6).

$N = 1500$: Number of Whisper encoder positions in a 30 s window (20 ms each).

T : Encoder positions covered by the utterance, $T = \lceil 50\tau \rceil$ for duration τ in seconds.

L, d, f : Number of encoder layers, model width and MLP width ($f = 4d$).

$\mathbf{X}^{(\ell)}, \mathbf{x}_j^{(\ell)}$: Hidden states of layer ℓ (before the final LayerNorm) and the state at position j .

$\mathbf{q}_i, \mathbf{k}_j, \mathbf{v}_j, a_{ij}$: Query, key, value and attention weight of a head.

$\bar{a}_j$: Received attention of position j , averaged over heads and queries.

$\Psi^{(\ell)}$: Ratio of the largest to the median absolute activation in layer ℓ .

$\mathbf{K}_{\emptyset}^{(\ell)}, \mathbf{V}_{\emptyset}^{(\ell)}$: Keys and values of layer ℓ for a 30 s silence input.

$\mathbf{S}$: Encoder outputs for the silence input; $\mathbf{S}_{T+1:N}$ is the “tail”.

$\mathcal{C}, \mathcal{D}$: Cached silence positions used as extra keys (encoder) and as extra memory (decoder).

$\mathbf{H}$: Encoder outputs given to the decoder.

$C(T_q, T_k)$: FLOPs of one encoder layer with T_q queries and T_k keys.

WER, Hal: Word error rate and hallucination rate (%).

Appendix B: Additional Derivations

a. Encoder cost. A pre-LN block applies four $d \times d$ projections (query, key, value, output) and a two-layer MLP of width f to each of the T_q query positions, i.e. $T_q(4d^2 + 2df)$ multiply-accumulates. Attention scores and the weighted sum of values each need $T_q T_k d$ multiply-accumulates. Counting a multiply-accumulate as two FLOPs gives

$$C(T_q, T_k) = 2T_q(4d^2 + 2df) + 2 \cdot 2T_q T_k d \quad (\text{B1})$$

$$= 2T_q(4d^2 + 2df) + 4T_q T_k d, \quad (\text{B2})$$

which is Eq. (6); the key/value projections for cached positions are computed once offline and are excluded. Layer norms, the convolutional stem (which processes only the $2T$ unpadded mel frames under SILENCECACHE) and the softmax are lower order. With a full cache every speech query sees exactly N keys (its own T plus the $N - T$ cached ones), so the relative cost of SILENCECACHE is

$$\rho_{\text{full}} = \frac{C(T, N)}{C(N, N)} = \frac{T [2(4d^2 + 2df) + 4Nd]}{N [2(4d^2 + 2df) + 4Nd]} = \frac{T}{N}. \quad (\text{B3})$$

The pilot’s mean T is 234.4 positions, so $\rho_{\text{full}} = 0.156$, matching the reported 0.157 (averaged per utterance). With a top- k cache the attention term shrinks to $4T(T+k)d$, and $C(T, T+k)/C(N, N)$ approaches the cost of unpadded encoding (0.11–0.14 in the pilot).

Appendix C: Implementation and Reproducibility Details

All code is in the project repository (`experiments/pilot_padding`). The encoder forward pass is written explicitly from the checkpoint weights and checked against the Hugging Face implementation (maximum absolute difference 6.9×10^{-4} at output scale 23.6 on a padded utterance; identical greedy transcript). Log-mel features use the official feature extractor without padding, with the waveform zero-extended to a multiple of 320 samples so that the number of encoder positions is an integer. Sinusoidal position embeddings are sliced to the first T positions. Cached keys and values are taken at positions $[T+1, N]$ of the silence run, so every speech query sees each absolute position at most once.

The full evaluation will add bootstrap confidence intervals (1,000 resamples), paired tests, wall-clock measurements with CUDA events and peak-memory reports, and will fix all selection rules before running on the test sets.

TABLE III. Software and settings of the pilot.

Item	Setting
Software	Python 3.14, PyTorch 2.11 (CUDA 12.8), Transformers 5.19, jiwer
Checkpoints	<code>openai/whisper-base</code> , <code>-small</code> , <code>-large-v3-turbo</code>
Precision	fp32 (base, small); fp16 (turbo)
Decoding	greedy; special tokens above end-of-text suppressed; 160 tokens max
Text normalisation	Whisper English normaliser (<code>tokenizer.normalize</code>)
Utterance selection	2–8 s test-clean utterances, shuffled with seed 0, first 100
Register frames	top- k token norm at the block after the emergence block
Sink frames	top- k received attention (all layers, heads, queries) in the silence run
Hardware	NVIDIA RTX 3050 Laptop GPU, 4 GB

Appendix D: Supplementary Results and Analyses

TABLE IV. Whisper-base with real partial zero padding and register transplants (pilot v1, same 100 utterances; WER %).

Condition	WER
30 s padding	5.39
no padding	385.46
1 / 2 / 5 / 10 s padding	31.20 / 16.34 / 5.94 / 5.86
no padding + tail from a <i>speech</i> reference	24.47
no padding + 4 transplanted register keys/values	315.40
both	25.02
padded; encoder attention to own 4 register frames masked	5.39

TABLE V. Top- k versus random- k cached silence positions (KV + full tail; WER %; pilot).

Model	top- k				random- k			
	4	16	64	256	4	16	64	256
Whisper-base	5.04	5.43	12.49*	5.20	5.66	5.74	5.51	5.20
Whisper-small	3.88	3.80	2.95	2.33	3.96	3.96	3.65	3.18
Whisper-large-v3-turbo	6.83	2.64	2.40	1.94	21.41	17.53	10.16	2.09

*Single outlier cell, to be re-checked at scale.

TABLE VI. Non-speech audio (ESC-50, 200 clips; pilot v3). Share of clips with any output word (%) and mean output words.

Condition	Whisper-base		Whisper-large-v3-turbo	
	any output	words	any output	words
clip + 25 s padding	1.0	0.80	78.5	1.10
clip only	8.5	1.37	98.0	16.27
clip only + silence tail	5.0	0.06	100.0	1.32
clip tiled to 30 s	3.0	1.05	94.0	2.81
clip tiled to 25 s + 5 s padding	4.5	1.60	87.5	6.30

These tables extend the main pilot with the same protocol. Additional robustness results (noise, long-form, beam search, languages) will be added from the full evaluation described in Sec. V.